Estimation Methods in NONMEM

ECP 8503: Intermediate Population PK/PD

Presented by Samuel Callisto, PhD

24 April 2026

University of Minnesota
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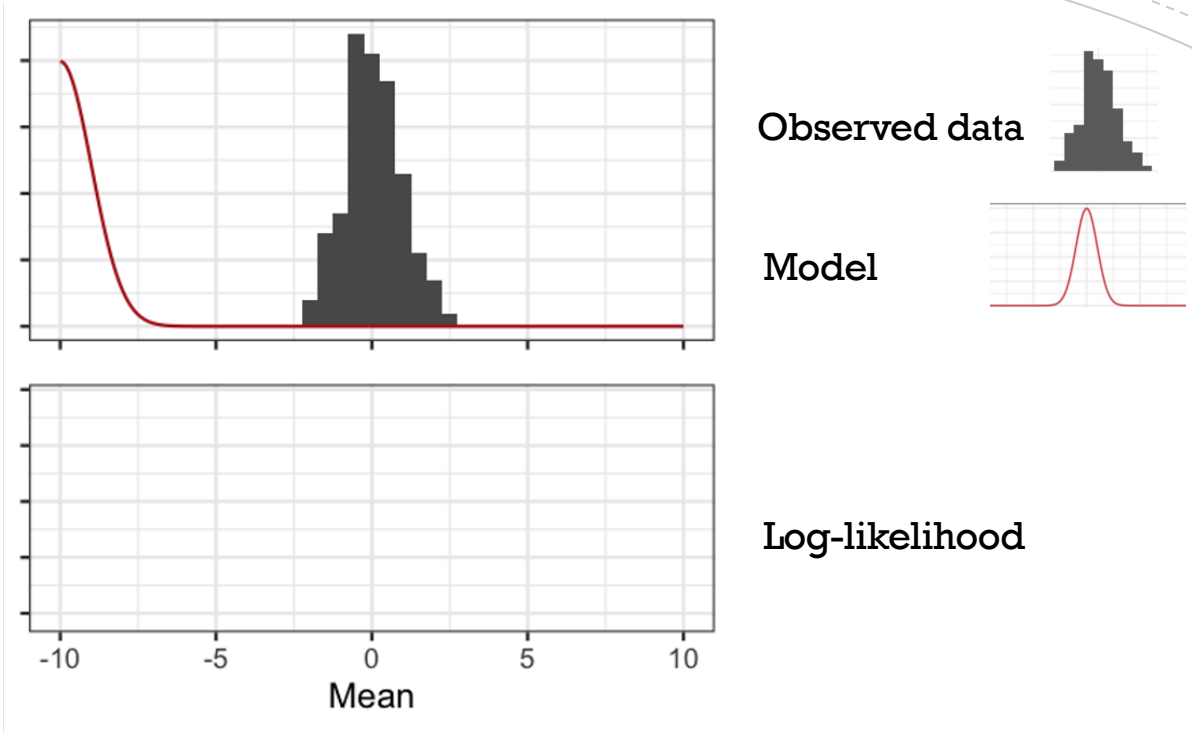
NONMEM Estimation Methods Discussed Today

- Maximum Likelihood Methods
  - FO
  - FOCE
  - LAPLACE
- Monte Carlo Expectation-Maximization (EM) Methods
  - SAEM
  - IMP
  - ITS
- Bayesian Markov Chain Monte Carlo Methods
  - BAYES
  - NUTS

How does NONMEM Work?

- Estimates the set of parameters (THETA, OMEGA, SIGMA) which best describe the population data
  - Considers ”all possible” combinations of individual parameters and ETAs (random effects)
- The maximum likelihood estimates (MLEs) of a set of parameters are <u>the values which maximize the likelihood</u>
  - Likelihood is the probability of the observed data considered as a function of the unknown parameters
  - NONMEM actually finds the <u>minimum</u> of -2*log-likelihood (-2LL) because it is more computationally efficient

Maximum likelihood estimation



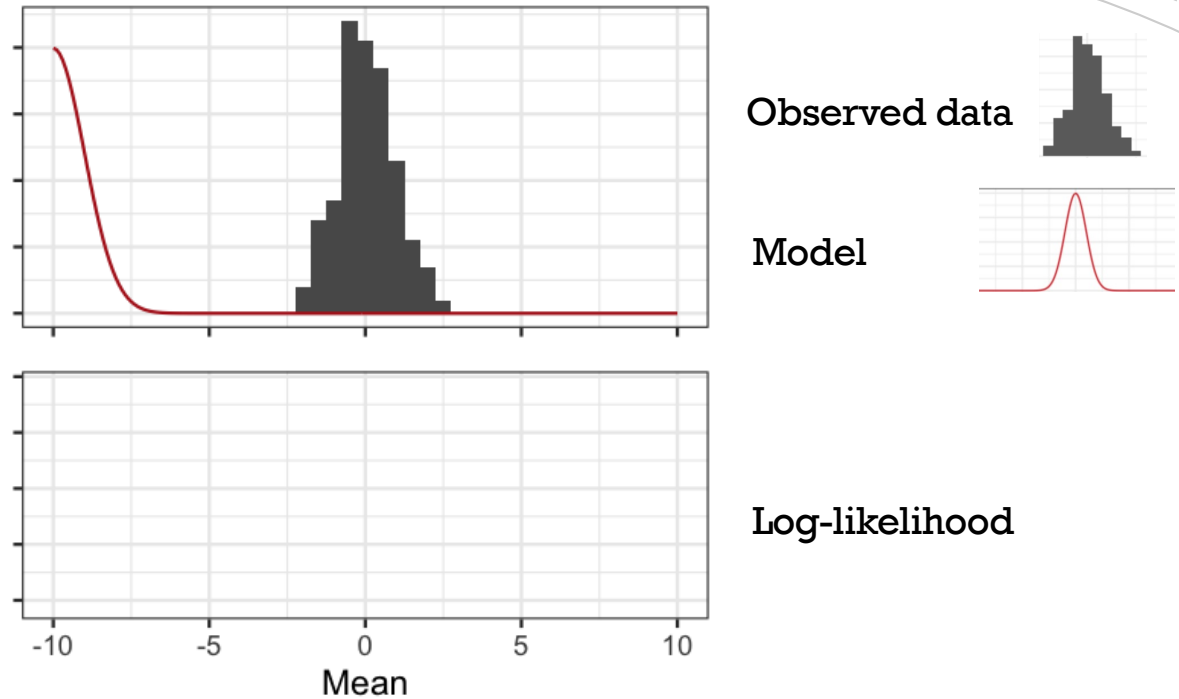
$$x_i = \mu + \varepsilon_i; \quad \varepsilon_i \sim N(0, \sigma = 1)$$

$$\log L \propto \sum_{i=1}^{n} -\frac{1}{2}\left(\frac{x_i - \mu}{\sigma}\right)^2$$

What value of μ will maximise $\log L$?

Animation courtesy of Dr. Tim Waterhouse

Maximum likelihood estimation



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Animation courtesy of Dr. Tim Waterhouse

How does NONMEM estimate models?

- Using a finite Taylor series, an approximation of a function estimated by a sum of terms that are calculated from the values of the function's derivatives at a single point
  - First order = first derivative
  - Second order = sum of first and second derivatives

<u>General form of First-Order Taylor Series</u>

$$f(x) \approx f(a) + \frac{f'(a)}{1!}(x - a)$$

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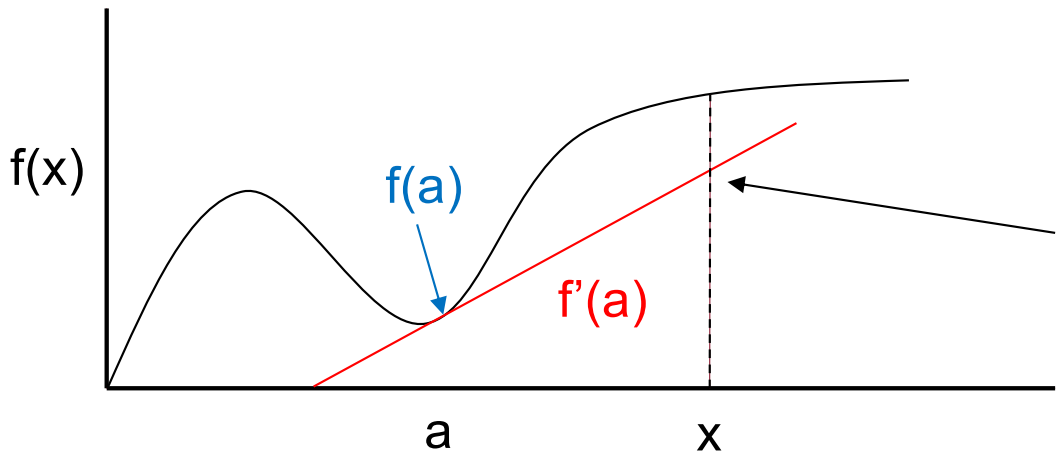


Where f'(a) intersects with x=x is the first-order approximation of f(x)

Why approximate in the first place?

- For non-linear mixed effects models there is <u>no closed-form solution</u> for the integral of the likelihood function over all possible ETAs
- This means we have to approximate the integral by linearizing the function (e.g., Taylor series), or by simulating it (EM methods)

First Order Approximation (FO)

```
$EST METHOD=0
```

- A first-order Taylor Series is used to approximate the function describing PK using your fixed and random effect parameters

$$
\begin{aligned}
Y_i &\approx g(\theta, x_i, \eta_0) + g'(\theta, x_i, \eta_0)(\eta_i - \eta_0) + \epsilon_i \\
&\approx g(\theta, x_i, 0) + g'(\theta, x_i, 0)(\eta_i) + \epsilon_i
\end{aligned}
$$

  - where $g(\theta, x_i, \eta_0)$ is a function describing the model, e.g. 2 compartment PK model
  - In FO, η_0 = 0 for all iterations of the ODE solver

FO Considerations

PROS

- Compared to other methods used in the 1970s, huge benefit of allowing correlation of observations within-subject
- Fast computation time

CONS

- Inaccurate estimates, especially in data sets with large residual error and/or large interindividual variability

FO: when should I use it

- If, and only if, you're running NONMEM on one of these

First-Order Conditional Estimate (FOCE)

```
$EST METHOD=1
```

- Still uses the first-order Taylor series to approximate the model

$$Y_i \approx g(\theta, x_i, \widehat{\eta_i}) + g'(\theta, x_i, \widehat{\eta_i})(\eta_i - \widehat{\eta_i}) + \epsilon_i$$

- In FOCE, η_0 is the mode of the posterior distribution for the previous iteration
  - Estimate is "conditioned on" the posterior mode
  - This means the Taylor series generates ETA estimates for each subject on each iteration

More details re: FOCE

- Integrates over all possible ETAs
  - Evaluates mode of the joint density (most likely values of ETAs)
  - Evaluate first order approximation of variances of ETAs
  - Integral is approximated using Gaussian function with this mean and variance
- Integration requires lots of computation time, which explains why FOCE runs slower than FO

When to use FOCE

- Most of the time this will be your default, especially during model development
- Data with rich and semi-rich sampling
- Good for models with many THETAs which have no ETAs associated with them

FOCE Considerations

PROS

- More accurate than FO/ITS
- Requires fewer adjustments than SAEM/IMP
- Reproducible up to ± 4 digits

CONS

- Doesn't handle full OMEGA blocks as easily as EM methods
- Requires dense data to estimate $\widehat{\eta_i}$

Eta-Epsilon Interaction (INTER)

errors are function of model

- $Y_{ij} = g(\theta, x_{ij}, \eta_i) \boxed{+ h(\theta, x_{ij}, \eta_i)\epsilon_{ij}}$
  - If you don't use the interaction option, η_i is set to zero when estimating ϵ_{ij}
  - There likely exists some interaction between our random effects that we want to account for when estimating our parameters
- "**<u>Always use INTER(ACTION) option</u>**. Without the INTERACTION option, the prediction F used in the multiplier expression to EPS() will be valuated at ETA = 0, which is contrary to how one should model data."[1]

1. Bauer, CPT-PSP. (2019) 8, 525–537

Second-Order Conditional Estimation (LAPLACE)

- `$EST METHOD=1 LAPLACE INTER`
- Named after French mathematician who found the solution to second-order linear partial differential equations
- Uses a <u>second-order</u> Taylor series to approximate the PK function
  - Similar to FOCE, η_0 is the mode of the posterior distribution for the previous iteration when using the LAPLACE method

Pierre-Simon Laplace
1749-1827

When to use the LAPLACE method

- Second-order Taylor series allows for non-Gaussian distributions of residual errors (`EPS`)
  - i.e. we are making no assumptions about Y when we use the Laplace approximation
- Use this method whenever you are modeling data that are not continuous from –Inf to Inf
  - Binary or Count data
  - Bounded continuous data
  - Categorical data

Originally added to NONMEM to model categorical pain relief data

LAPLACE Considerations

PROS

- Allow for non-continuous data to be modeled

CONS

- Will not give good estimates when the posterior distribution of ETAs is not symmetrical
- Longer run times due to additional computation of second derivative

Monte Carlo Expectation-Maximization

- Set of stochastic algorithms which do not use maximum likelihood to estimate the set of parameters in your model
- Iterative process in which each iteration consists of two steps
  - Expectation step
  - Maximization step

The EM Algorithm

- Expectation (E) step
  - Monte Carlo integration (i.e., random sampling of parameters) performed to explore entire ETA space
  - calculate expected value of likelihood, given current parameter estimates
- Maximization (M) step
  - find parameters that maximize the likelihood from the E step

What's Monte Carlo Integration?

- Numerical evaluation of an integral
  - The integral of interest here being the fixed effect estimates for each subject
- A random sample of n values of x are drawn from the distribution of $f(x)$
- The sample mean is then used to estimate the integral of $f(x)$

General EM Considerations

- Runs slower than FOCE on simple models
  - E.g. less than six parameters, no ODEs
- Full OMEGA blocks are preferred in EM methods because they are easier to update ("maximize") than diagonal
- More options to adjust than in FOCE
- Stochastic algorithms will not produce the same answer every time, typically varying by ~25% of the SE

Importance Sampling

- `$EST METHOD=IMP INTER SEED=123 MAX=9999 EONLY=0 NITER=300 ISAMPLE=1000 RANMETHOD=P NOABORT`
- During expectation step, randomly samples ETAs for each subject and evaluates conditional density
  - Random samples taken from a normal distribution approximating the conditional density
  - "Important" region of conditional density
- Uses conditional Monte Carlo mean and variance from previous iteration

Importance Sampling with MAP

- `$EST METHOD=IMPMAP`
- Use the Importance Sampling method assisted by Mode a Posteriori (MAP) estimation
- At each iteration, conditional modes and conditional first-order variances are evaluated as in FOCE
- These are then used as parameters to the multivariate normal proposal density for the Monte Carlo importance sampling step

When to Use IMP/IMPMAP

- Good for complex models with ODEs and/or many parameters
- Rich or sparse data (fewer data points per subject than ETAs to be estimated)

IMP/IMPMAP Considerations

PROS

- Can track progress of improvement in OFV with each iteration
- Can handle full OMEGA blocks

CONS

- Can be less accurate than SAEM with highly categorical or sparse data
- Results can vary stochastically, typically by about 25% of SE
- Less efficient when not all THETAs are MU referenced

What's MU referencing?

An aside about MU-referencing. . .

- "THETAs that define typical values of individual parameters are 'mu related'"[1]
  - $\phi_j = \mu_j + \eta_j$, where μ_j is a function of θ_k
- "Fixed effect THETAs that have no intersubject variability associated with them but served as a shared value among subjects are not mu related"[1]
- Simplifies updating of THETAs during maximization step
  - ϕ_j is normally distributed & only in second term of integral
  - FOCE more efficient than EM if >3 THETAs are not MU referenced

1. Bauer, CPT-PSP. (2019) 8, 525–537

NONMEM Code for MU Referencing

$$e^{\theta_2 * \log\left(\frac{AGE}{50}\right)} = \frac{AGE^{\theta_2}}{50}$$

```
$PK
CLAGE = THETA(2)*LOG(AGE/50)
CLSEX = SEX*THETA(3)
CLWT = (WT/70)**THETA(4)
LTVCL= THETA(1)+CLAGE+CLSEX
MU_1= LTVCL
CL=CLWT*EXP(MU_1+ETA(1))
```

TH1/2/3 estimated on log scale

Time-varying covariates cannot be mu-referenced

MU_n & ETA(n)

Stochastic Approximation Expectation Maximization

- `$EST METHOD=SAEM INTER NBURN=2000 NITER=1000 NOABORT MAXEVAL=9999 RANMETHOD=P SEED=123`
- Randomly sample ETAs from proposal density for each subject, centered at the previous sample position
- New samples accepted with probability related to conditional density
- Proposal density variance is adjusted to maintain average acceptance rate
- Two phases:
  - Stochastic "burn-in" phase: approximates individual parameters; update population parameters from individual parameters
  - Accumulation phase: average together individual parameter samples from previous iterations

Obtaining OFV from SAEM

```
$EST METHOD=SAEM AUTO=1 NITER=500
$EST METHOD=IMP EONLY=1 NITER=5 ISAMPLE=1000 MAPITER=0
```

- The objective function estimates from SAEM should not be interpreted as a measure of model fit
- To obtain an interpretable OFV, we run an <u>expectation-only</u> IMP with a few (1-10) iterations after SAEM to get an accurate estimate of OFV
- You will use the <u>final estimates from IMP</u> for both OFV and parameter estimates

When to Use SAEM

- With categorical data as an alternative to LAPLACE
- Good for complex PKPD models or models with many parameters (including full OMEGA blocks)
- First alternative to try if you are having trouble fitting model with FOCE-I (Sam's recommendation)

SAEM Considerations

PROS

- Handles very sparse to rich data easily
  - Better than IMP for sparse!
- Can handle categorical data
- Can handle full OMEGA blocks

CONS

- Doesn't produce interpretable objective functions!
- Results can vary stochastically, typically by about 25% of SE
- Less efficient when not all THETAs are MU referenced

Iterative Two-Stage (ITS)

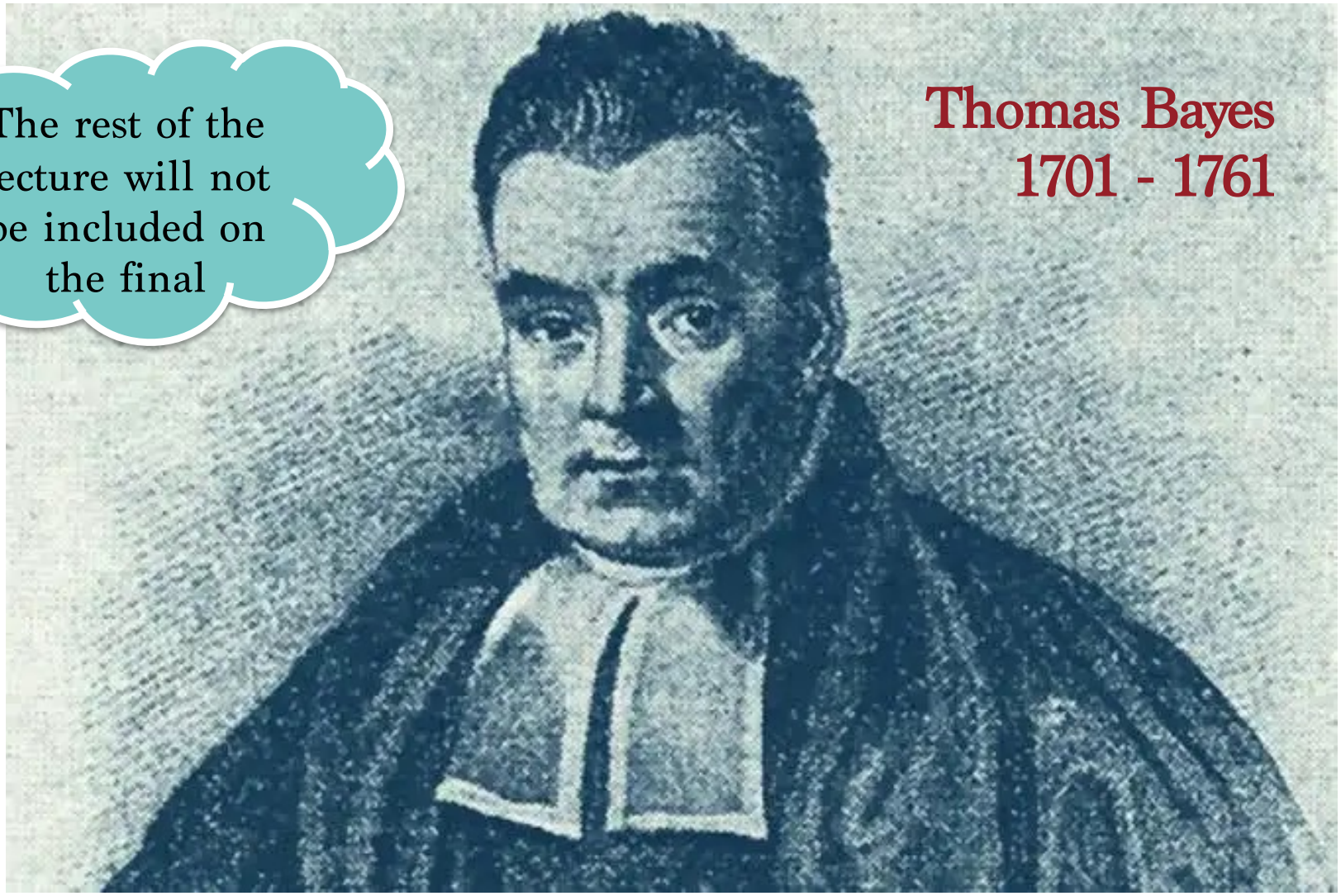
- Approximate expectation-maximization algorithm
- Typically used as a preliminary analysis to improve parameter estimates for EM/Bayesian methods[1]

```
$EST METHOD=ITS INTER AUTO=1 NOABORT
$EST METHOD=SAEM AUTO=1 NITER=500
$EST METHOD=IMP EONLY=1 NITER=5 ISAMPLE=1000
     MAPITER=0
```

NOTE: If you use chains of EST like this, you want to use the estimates from the FINAL estimation step when you report your results!

1. Bauer, CPT-PSP. (2019) 8, 525–537

The rest of the lecture will not be included on the final



Key Messages

- Bayesian statistical approaches are fundamentally different from classical, frequentist approaches
- Additional diagnostic plots and numerical calculations are required to interrogate model outputs
- You can't trust NONMEM-generated outputs for Bayes runs, instead you need to do post-processing of model outputs to get actual population and individual parameter estimates

A Simple Description of Bayesian Statistics

Count all the ways that data can happen according to your assumptions. Outcomes which are consistent with the data and occur most frequently are the most plausible.

-- Paraphrased from Richard McElreath

Bayes: A Paradigm Shift

- Frequentist statistical approaches seek to obtain a point estimate for each parameter
- Bayesian statistics provides us with a distribution of plausible values for each parameter
  - Prior distributions define our knowledge of a parameter before fitting to data
  - Posterior distributions are the updated view of a parameter that we have after fitting a Bayes model to data

A Simple Bayes Model

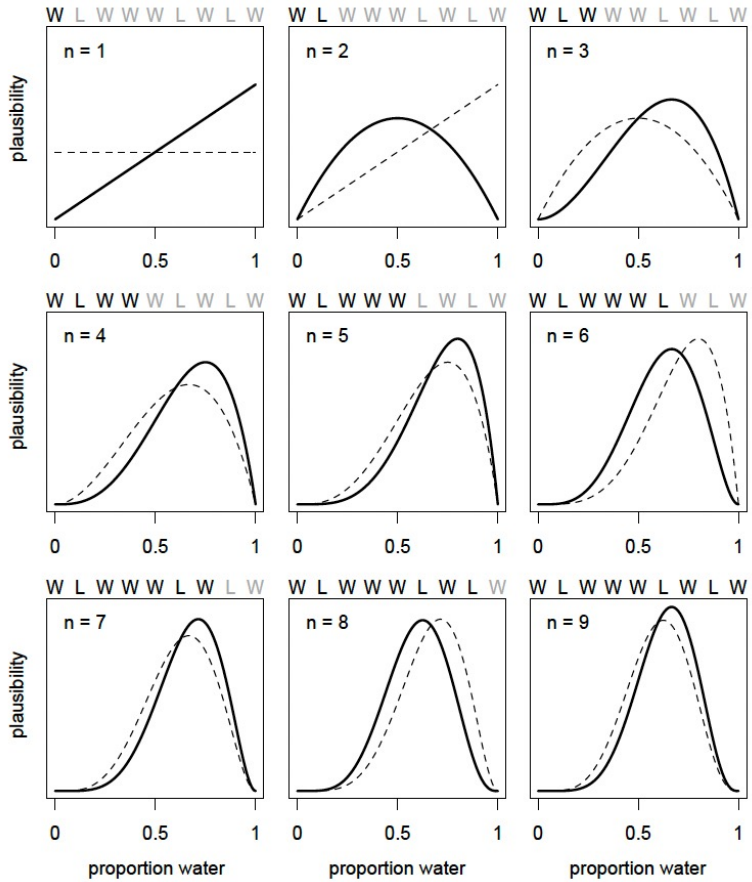


FIGURE 2.5. How a Bayesian model learns. Each toss of the globe produces an observation of water (W) or land (L). The model's estimate of the proportion of water on the globe is a plausibility for every possible value. The lines and curves in this figure are these collections of plausibilities. In each plot, previous plausibilities (dashed curve) are updated in light of the latest observation to produce a new set of plausibilities (solid curve).

Dashed = prior; solid = posterior

Figure copied from Statistical Rethinking 2e; 2020.

Bayes' Theorem

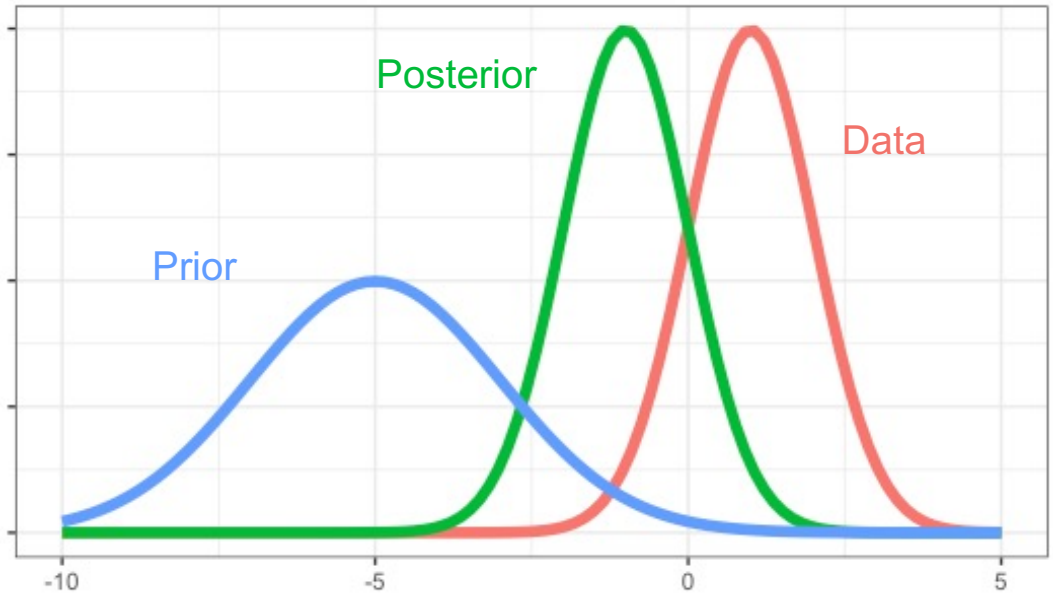
$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

- A = subject has Covid-19, B = positive PCR test result
- P(B|A) = 0.80 (Sensitivity; 20% false negative)[1]
- P(!B|!A) = 0.99 (Specificity; 1% false positive)[1]
- P(A) = 0.0027 (0.27% prevalence Hennepin County 12/14/21)[2]
- P(B) = P(B|A)P(A) + [1-P(!B|!A)]P(!A)
- P(A|B) = 17.8%

1. https://www.cap.org/member-resources/articles/how-good-are-covid-19-sars-cov-2-diagnostic-pcr-tests 2. https://www.nytimes.com/interactive/2021/us/hennepin-minnesota-covid-cases.html

Bayes' Theorem in PMx

$P(\theta, \Omega$



$, \Omega, \Sigma)$

$$Posterior\ Distribution = \frac{Likelihood\ *\ Prior\ Distribution}{Average\ Probability\ of\ the\ Data}$$

Markov Chain Monte Carlo (MCMC)

- Several techniques exist to numerically predict the posterior distribution
  - E.g. grid approximation or quadratic approximation
- MCMC does not attempt to approximate posterior, instead samples are randomly drawn directly from the posterior
- This results in a collection of parameter values, the frequency of which corresponds to the posterior plausibility of that parameter value

Sampling from the Posterior

- How can we sample from the posterior if we don't know it? Several algorithms have been developed to do this
- "Guess and Check" Algorithms
  - Metropolis (original)
  - Metropolis-Hastings (more generalized)
  - Gibbs sampling (efficient version of M-H)
- Physics Simulation
  - Hamiltonian Monte Carlo (currently the most commonly used approach)

M-H and Gibbs

WinBUGS: Bayesian inference
Using Gibbs Sampling

- M-H: a set of parameter estimates is randomly selected. Samples are accepted at some rate, calculated using a ratio of a function using the current and proposal parameters. If accepted, proposal parameters used in next iteration, otherwise the current parameters are kept (Markov Chain)
- Gibbs: Distribution of proposed parameter values adjusts depending upon the current parameter values, thus leading to more efficient sampling (efficiency = better estimates from fewer iterations)

What's No U-Turn Sampler (NUTS)?

- An implementation of Hamiltonian Monte Carlo in NONMEM
- A particle is launched across the sampling space and rolls according to the probability gradient. NONMEM is actually doing a simulation of a physical representation of this phenomenon
- Although more computationally difficult, the acceptance rate is significantly higher than M-H or Gibbs sampling
- The U-Turn problem: samples get stuck in same position. Leads to high autocorrelation between iterations

What the Posterior Distribution Tells Us

- MCMC output is a posterior distribution of parameter estimates, rather than a single value (point estimate)
- Instead of estimating confidence intervals based on the asymptotic standard error, we instead look at the quantiles of the posterior distribution
  - This method makes no assumption about the shape of the posterior distribution, and will allow for asymmetric intervals
- This approach may result in a faster runtime than if you had run a bootstrap, which approximates the posterior

Bayesian Estimation in NONMEM

```
$EST METHOD=BAYES INTERACTION
```

- Full Markov Chain Monte Carlo Bayesian Analysis Method
- Iterations are split into two stages similar to SAEM
  - (1) Burn-in and (2) Stationary Distribution
  - Typically at least 10,000 iterations are performed in the stationary distribution phase
- Also like SAEM, the OFV is not interpretable as a likelihood integrated over all possible etas
  - Instead, it is the likelihood at the set of etas selected during that iteration for both burn-in and stationary distribution phases

Two Sampling Phases in Bayes

- Burn-in Phase
  - highly stochastic sampling of parameter space
- Stationary Distribution or Accumulation Phase
  - conducts more constrained search of defined space
  - estimates vary randomly with each iteration, without changing on average
  - The posterior is the aggregate of all of the samples taken during the accumulation phase

Why MU Referencing is Importa

Gibbs is Good
M-H is MeH

- If parameters are MU referenced as a linear function of THETA, NONMEM can use the Gibbs sampling method

```
EM/BAYES SETUP:
THETAS THAT ARE MU MODELED:
1 5 12 13 14 15
THETAS THAT ARE GIBBS SAMPLED:
1 5 12 13 14 15
THETAS THAT ARE METROPOLIS-HASTINGS SAMPLED:
2 3 4 6 11
```

THETAs 7-10 are fixed allometric constants in this model, so no sampling occurs

Defining the Prior Distribution

```
$PRIOR NWPRI NTHETA=10 NTHP=6 NETA=2 NETP=2
```

- You DO NOT need a prior for all thetas/etas
- You DO NOT want to provide a prior for epsilons
- Numeric order matters here! Parameters with priors come first, those without come last
  - E.g. in this $PRIOR statement above, THETAs 1-6 have priors, but THETAs 7-10 do not. Both ETAs have priors defined

Defining Priors for Thetas

```
$THETAP
5.4 FIX ; CL
27 FIX  ; V
```

Prior distribution mean

```
$THETAPV BLOCK(2)
1 FIX ; 1 - CL
-0.76  1 ; 2 - V2
```

Prior distribution variance

- This defines how "informative" our prior is

Prior Predictive Check

```
summary(exp(rnorm(10000,5.4,0.1)))
```

- Take 10,000 random draws from normal distribution with mean=5.4 and SD=0.1 <u>on the log scale</u>
- This summary will show you the expected range of values that your prior would predict
- Verify that you aren't including physiologically implausible values (e.g. THETAPV=10 is huge!)

Defining Priors for Omegas

NONMEM will have a hard time fitting without OMEGA priors

- Because we used NWPRIOR, our omega priors follow a Wishart distribution

```
$OMEGAP BLOCK(2)
0.09 FIX      ; 1 - CL
0.04  0.09    ; 2 - V

$OMEGAPD
(2 FIX)
```

"Scale matrix": Prior estimate of variance-covariance matrix

Degrees of freedom of omega prior

- This defines informativeness
- Totally uninformative = OMEGAP block size

Defining Priors for Sigmas

DON'T

- This would assume your errors follow similar distribution
- In most cases, this is a bad assumption

Avoiding Autocorrelation Using Chains

- Within a single model run, because we are using an MCMC estimation method there may be correlation between sequential iterations ("autocorrelation")
- To avoid this, we run several separate models (”chains”) in parallel, then combine their results at the end of the run
- Chains can differ in their initial estimates or in the seed provided to the random number generator

$EST METHOD=CHAIN

- Built-in NONMEM function that generates an independent set of initial estimates for each chain based on the provided prior distributions
- Separate model files can then be automatically created using these sets of parameter estimates
- You should always run multiple chains for Bayes runs, and this is an easy way to generate the chain files

”Knobs to Turn” in $EST

- NBURN=400
  - Number of burn-in iterations. When all parameters/OFV appear to bounce around in a stationary region, they are “burned” in.
- NSAMPLE/NITER=10000
  - Number of stationary distribution iterations to perform
- THIN=1
  - Bayesian records retained in the raw output file may be adjusted by every THINth iteration

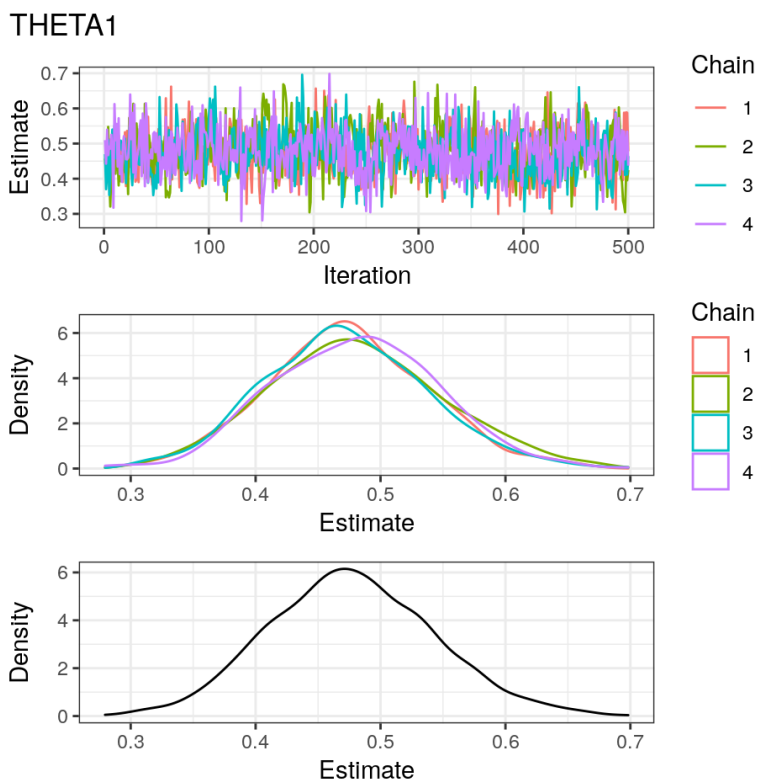
. . . and many more! Check out the User’s Guide

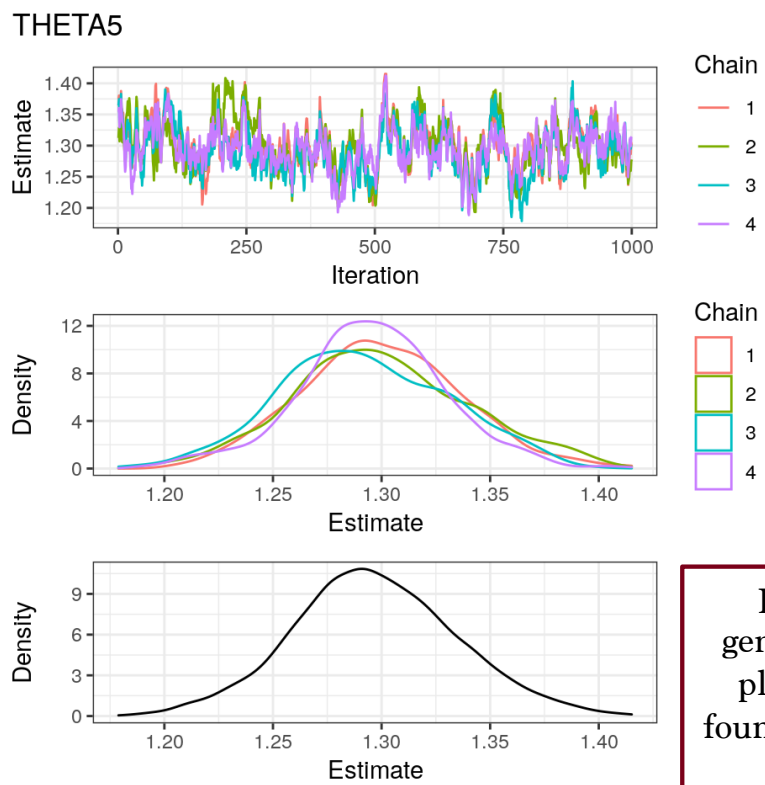
Model Evaluation

Visually Monitoring the Model

- Trace Plots
  - Check for autocorrelation in the accumulation phase
  - Ideally these plots with overlaid chains will look like a ”fuzzy caterpillar” (yes, that’s the technical term)
- Visualization of Posterior
  - Distribution shape should match what you were expecting
  - Should see general agreement between chains (but not identical)

Examples of Good and Bad Diagnostics





R code to generate these plots can be found in backup slides

Numerical Evaluation of Model Convergence

- Gelman-Rubin Diagnostic ($\hat{R}$ or "R-hat")
  - Ratio of between-chain variability to within-chain variability
  - Ideally $\hat{R}$ =1.00, with a value of 1.02 on the edge of acceptable
  - High values ($\hat{R}$ =1.05) indicate lack of convergence
- Effective Sample Size ("n-eff" or "ESS")
  - The number of independent samples achieved for this parameter
  - Generally want at least 100 per parameter; less than that indicates high autocorrelation between iterations[1]
  - Separate ESS for Bulk (location, e.g. mean/median) and Tail (5th/95th percentiles) of posterior distribution

1. https://mc-stan.org/rstan/reference/monitor.html

Generating $\hat{R}$ and n-eff

- In R we can use the `rstan` package to generate these values
- `rstan::monitor(sims)`
  - Where `sims` is a 3D array (iterations*chains*parameters) from an MCMC model (doesn't matter which software was used, e.g. Stan or NONMEM)
  - This is quicker than generating these statistics yourself (and less prone to errors as long as you get the structure of `sims` correct)
  - Check out vignettes online for how to run this function

If you only hear one thing today, hear this:

- When NONMEM reports model results, it outputs the parameter estimates from the final iteration
- With Bayes, each iteration is a sample of parameters from the posterior distribution, so we do not want to look at estimates from a single iteration
  - **This means we can't trust population parameter estimates, tabled EBEs or ETAs, or even the shrinkage estimates reported in the default NONMEM output ($TABLE)**
- Instead, the posterior must be post-processed to produce the correct parameter estimates from the model run

Using the Posterior

- Rather than point estimates for model parameters, Bayes models output a posterior distribution
- When reporting model results, we need to calculate values using the posterior, rather than just using the values reported in the LST file
  - This is because the NONMEM outputs are from the final iteration, rather than summarizing the entire posterior

Don't Include `$COVARIANCE`

- Do not conduct a `$COVARIANCE` step when running Bayes! You will not use the outputs for anything, and it will significantly increase your runtimes
- `$COVARIANCE` estimates the standard error about your parameters using the variance-covariance matrix
- In Bayes, we can directly calculate the standard error from the posterior distribution for each parameter

Comparison with Bootstrap Approach

	BAYES	Bootstrap
Parameter Estimate	Median of the entire posterior	Median of set of simulations
95% Interval	Quantiles of the posterior; referred to as “credible intervals”	Quantiles of the vector of estimates from simulations
Model Stability	ESS and R-hat	% of completed runs

Example Parameter Table

			Median	95% CDI	Bulk ESS	Tail ESS	$\hat{R}$
CL/F (L/h)	$\exp(\theta_1)$	Apparent clearance	10.3	(9.12, 10.8)	4919	15222	1.00
Vc/F (L)	$\exp(\theta_2)$	Apparent central volume	4.21	(3.85, 5.12)	874	1823	1.00
KA (1/h)	$\exp(\theta_3)$	First order absorption rate constant	0.369	(0.355, 0.378)	714	1715	1.00
Q/F (L/h)	$\exp(\theta_4)$	Apparent intercompartmental clearance	4.98	(4.31, 5.21)	657	1536	1.00
Vp/F (L)	$\exp(\theta_5)$	Apparent peripheral volume	68.3	(59.6, 79.2)	471	1177	1.01
D1 (h)	$\exp(\theta_6)$	Zero order absorption duration	0.248	(0.166, 0.411)	1027	1665	1.00

The model used mu-referencing: estimates presented here were back-transformed from the log-domain for clarity
Abbreviations: CDI = credible interval; ESS = effective sample size; $\hat{R}$ = Gelman-Rubin diagnostic
Credible intervals calculated from Bayesian posteriors
Source code: 09-Bayes-Parameter-Tables.R
Source file: 142_param_struct.tex

Bayes Tips & Tricks

When to Use Bayes

- Useful for evaluating parameter confidence intervals after model development with other methods
- Sampling directly from posterior can aid with simulations involving parameter uncertainty
- Can fit any type of data (continuous, bounded, categorical)
- Runtimes can be long depending on model complexity, so having access to multicore processors is helpful

Troubleshooting Bayes Runs

- My trace plots show high autocorrelation
  - Add more burn-in iterations
  - Reduce number of estimated parameters in your model
- My density plots look weird
  - Add more burn-in/sampling iterations
  - Constrain your prior distribution for problem parameters
  - Run ITS as a first estimation step to improve initial estimates

Troubleshooting Bayes cont.

- My R-hat is >1.02
  - Reparameterize the offending parameter
  - Run ITS as a first estimation step to improve initial estimates
  - Add more sampling iterations
- My model isn't converging
  - Constrain bounds of priors
  - Switch to NUTS

Conclusions

- Outputs from Bayesian statistical approaches are the posterior parameter distributions
- Model runs should be investigated using trace plots, parameter density plots, R-hat, and n-eff
- NONMEM outputs must be post-processed to get actual parameter estimates from the model run
- DON'T USE THE VALUES REPORTED IN THE LST FILE!

References/Further Reading

Beal, S.L., Sheiner, L.B., Boeckmann, A.J. & Bauer, R.J. (Eds). NONMEM 7.4 users guides (ICON plc, Gaithersburg, MD, 1989–2018). <https://nonmem.iconplc.com/nonmem743/guides>.

Bauer RJ. NONMEM Tutorial Part II: Estimation Methods and Advanced Examples. *CPT Pharmacometrics Syst Pharmacol.* 2019;8(8):538-556. doi:10.1002/psp4.12422

Bach T, An G. Comparing the performance of first-order conditional estimation (FOCE) and different expectation-maximization (EM) methods in NONMEM: real data experience with complex nonlinear parent-metabolite pharmacokinetic model. *J Pharmacokinet Pharmacodyn.* 2021 Aug;48(4):581-595. doi: 10.1007/s10928-021-09753-0. Epub 2021 Apr 21. PMID: 33884580.

Notes from Adrian Dunne's TACA Training Course (2018)

Selected Bayes References

Video explanation of several Bayes algorithms:

- https://chi-feng.github.io/mcmc-demo/

Bauer RJ. NONMEM Users Guide – Introduction to NONMEM 7.4.3. May 17, 2019.

McElreath R. (2016). *Statistical rethinking: a Bayesian course with examples in R and Stan.* CRC Press, Taylor & Francis Group.

Dr. McElreath's Statistical Rethinking course is available free online

First lecture: https://www.youtube.com/watch?v=4WVelCswXo4

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$EST METHOD=1 INTER FAST

alternate gradient calculation

FAST (NM74)

The FAST option is available for FOCE/ITS methods. The FAST method allows use of analytical theta derivatives to facilitate FOCE analysis. All thetas should be MU-referenced in the manner described in Guide INTRODUCTION TO NONMEM 7, "MU Referencing". For thetas that should not have inter-subject variability, or should not be MU referenced, MU reference it anyway by adding additional etas and assigning them to these thetas through MU referencing, but set their associated omega values to 0.0 FIXED.[1]

1. NONMEM Help Guide, $ESTM

R Code to Import the Posterior

```
library(tidyverse)

df <- map_df(1:4, ~ data.table::fread(
          glue::glue("model/101/101_{.x}.ext")
      ) %>% mutate(chain=.x)
) %>% filter(ITERATION>0)
```

R Code to Generate Trace Plots

```
## (run code to import posterior before this)
df.long <- df %>%
  pivot_longer(-c(ITERATION, chain))

map(unique(df.long$name), ~ df.long  %>%
    filter(name == .x) %>%
  ggplot(aes(x=ITERATION, y=value,
         colour = factor(chain))) +
    geom_line() + labs(y=.x)
)
```

ITERATION	chain	name	value
1	1	THETA1	1.71
1	1	THETA2	2.06
1	1	THETA3	-1.55
1	1	THETA4	0.99

R Code to Generate Posterior Density Plots

```
## (run code in previous slides before this)

map(unique(df.long$name), ~ {
  p2 <- ggplot(filter(df.long , name == .x)) +
    geom_density(aes(value, colour = factor(chain))) +
    labs(x = "Estimate", y = "Density", colour = "Chain")
  p3 <- ggplot(filter(df.long , name == .x)) +
    geom_density(aes(x = value)) +
    labs(x = "Estimate", y = "Density")
  return(list(p2, p3))
})
```

Post-Processing the Posterior R Code

```
## (run code to import posterior before this)

par_med <- df %>% summarize_all(~ median(.x) %>% round(3))
par_lo <- df %>%
  summarize_all(~ quantile(.x, prob = 0.025, names=FALSE) %>%
                  round(3))
par_hi <- df %>%
  summarize_all(~ quantile(.x, prob = 0.975, names=FALSE) %>%
                  round(3))

glue::glue("{par_med} [{par_lo},{par_hi}]")
```